

TrES-5b

R-0918

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_250716 · created 2026-07-29T08:22:49+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460872.7194 +/- 0.0043 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.099 +/- 0.054
Transit depth (Rp/Rs)^2	0.99 +/- 1.08 [%]
Orbital Inclination (inc)	80.24 +/- 2.86
Airmass coefficient 1 (a1)	0.9 +/- 0.5
Airmass coefficient 2 (a2)	0.19 +/- 0.9
Scatter in the residuals of the lightcurve fit is	53.03 %
Best Comparison Star	#3 - [604, 255]
Optimal Aperture	4.11
Optimal Annulus	14.51
Transit Duration (day)	0.03 +/- 0.029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.099 $\pm$ 0.054 vs 0.142 (deviation 0.8 σ)
Duration fit vs published	0.03 d vs 0.0758 d (deviation 1.58 σ)
Mid-time O-C	-4.5 min
Depth SNR	1.8
Residual scatter	53.03 %

Light curve

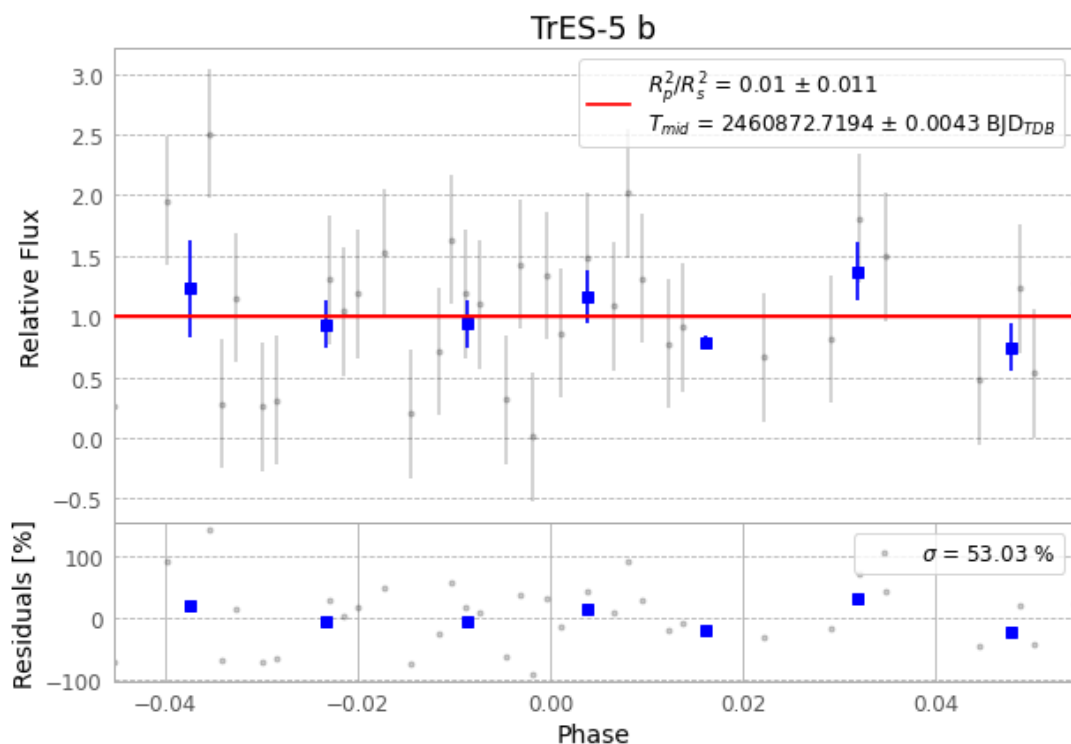


Figure 1 — FinalLightCurve_TrES-5 b_2025-07-15.png (SHA-256 35b72f3897f5b466...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [490, 374], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 20566a33e137a23e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

72 FITS frames (47 MB), TRES-5250716033615.FITS → TRES-5250716070916.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-250716.log (307178733e2d...), FinalLightCurve_TrES-5 b_2025-07-15.png (35b72f3897f5...), FinalLightCurve_TrES-5 b_2025-07-15.csv (e4a669bad82c...)

Compute resources

Wall time 130.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 08:22Z.